Sales Data Analysis Report

1. Total Revenue

This analysis begins by calculating the **total revenue generated by the company** during the analyzed period.

The results show that the company produced **strong overall revenue**, indicating consistent sales activity.

```
Total_Revenue = df["Revenue"].sum()

print("The total revenue is:", round(Total_Revenue,2))
✓ 0.0s
The total revenue is: 926800
```

2. Average Revenue per Transaction

The average revenue per transaction measures how much money the company earns **from each individual sale**.

This KPI helps evaluate:

- Customer spending behavior
- Pricing strategy effectiveness
- Overall transaction value

```
Avrage_order_value = df["Revenue"].mean()
print("Average Order Value:", round(Avrage_order_value,2))
```

0.0s

```
Average Order Value: 1853.6
```

3. Top Revenue-Generating Products

A product-level analysis was conducted to determine **which products generate the highest revenue.**

Understanding top-performing products helps the company:

- Focus marketing efforts
- Improve inventory management
- Maximize profitability

```
Revenue_by_product =(
    df.groupby("Product")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)
print(Revenue_by_product)
```

0.0s

```
Product
Laptop        153350
Monitor       150550
Mouse         133900
Tablet        132700
Headphones    127350
Keyboard      124100
Smartphone    104850
Name: Revenue, dtype: int64
```

4. Revenue by Category

Revenue was also analyzed across **product categories** to identify which groups of products contribute the most to total sales.

This analysis helps the business understand:

- Which product groups perform best
- Where demand is strongest
- Which categories may require improvement

```
Revenue_by_category =(
    df.groupby("Category")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)
print(Revenue_by_category)
```

[8] ✓ 0.0s

```
Category
Electronics    668800
Accessories    258000
Name: Revenue, dtype: int64
```

5. Revenue by Region

Regional analysis examines how revenue is distributed **across different geographic areas**.

This helps identify:

- High-performing markets
- Areas with growth potential
- Regional differences in customer demand

```
Revenue_by_region =(
    df.groupby("Region")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)
print(Revenue_by_region)
```

```
[10]  ✓ 0.0s

Region
West       231050
North      184550
East       180300
Central    175500
South      155400
Name: Revenue, dtype: int64
```

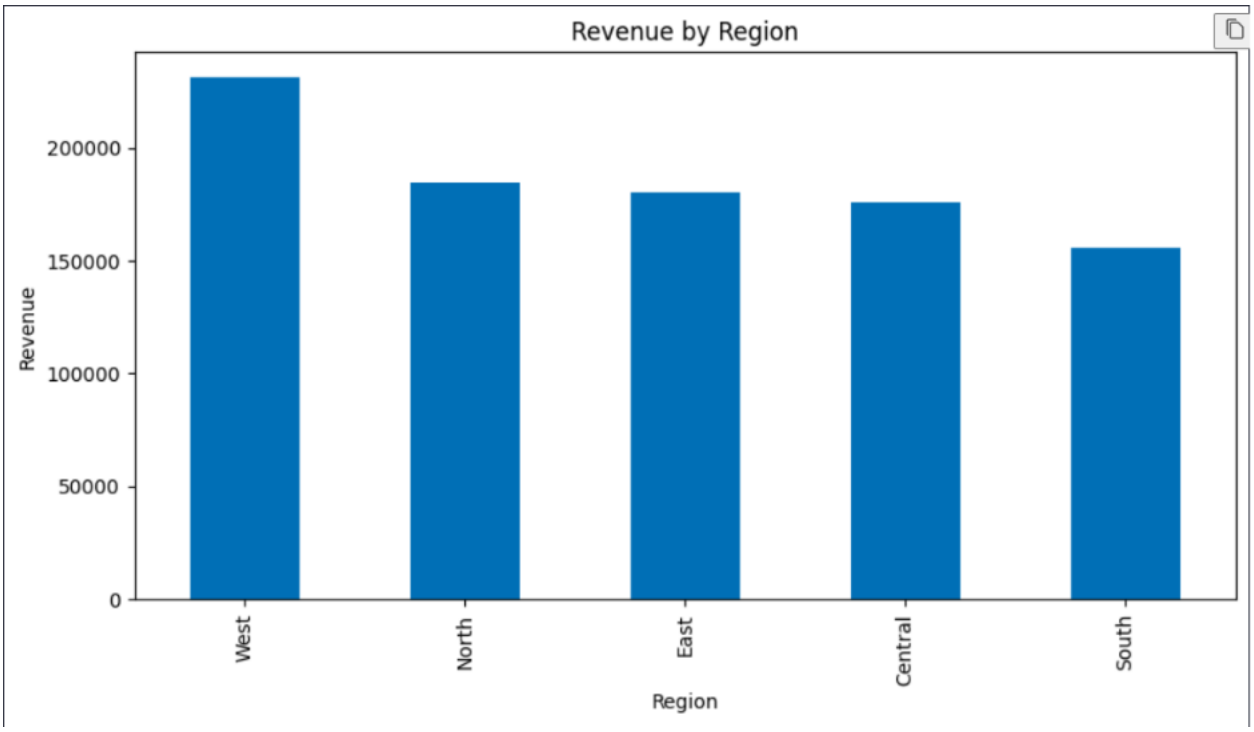
6. Regional Revenue Visualization

A regional revenue chart was created to visually compare **sales performance across regions**.

This chart allows stakeholders to quickly identify:

- The strongest markets
- Regions with lower sales performance

```
Revenue_by_region.plot(
    kind="bar",
    figsize=(10,5),
    title="Revenue by Region"
)

plt.xlabel("Region")
plt.ylabel("Revenue")

plt.show()
```



7. Revenue Calculation

Revenue was calculated using the following formula:

Revenue = Quantity × Price

This calculation provides the foundation for all further analysis within the report.

```
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")

df["Month"] = df["Date"].dt.to_period("M")

Montly_revenue = df.groupby("Month")["Revenue"].sum()
print(Montly_revenue)
```

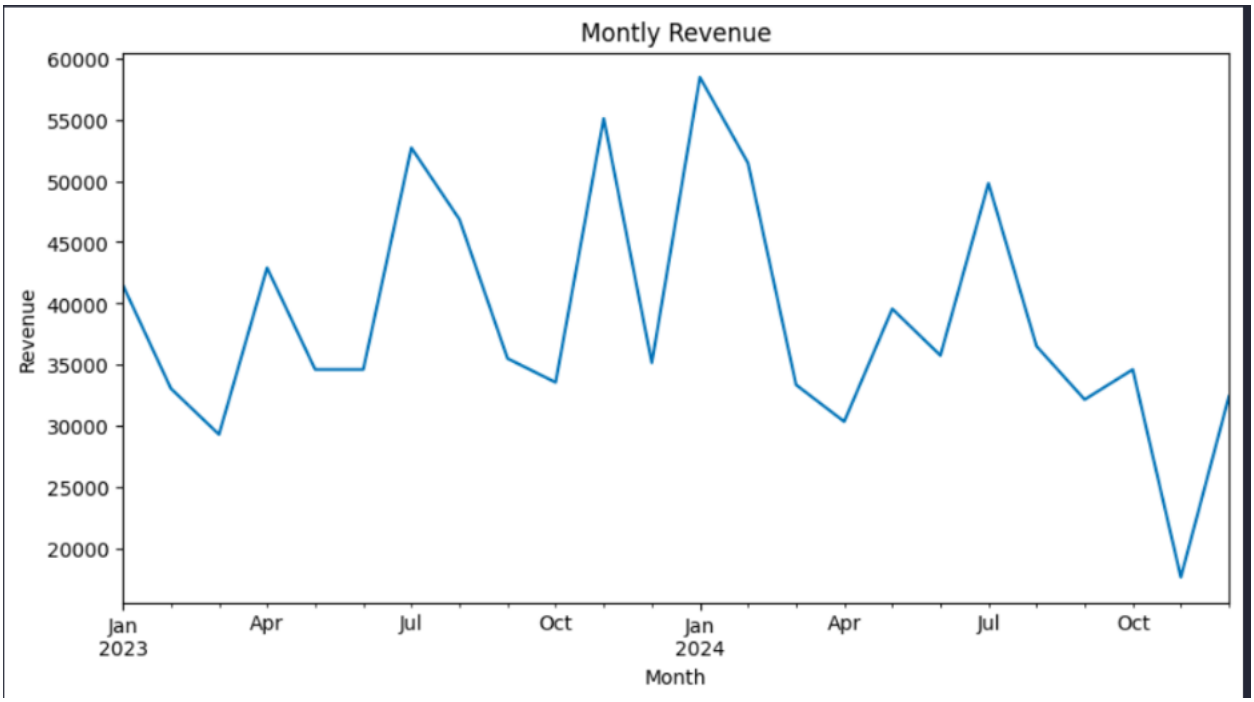
```
Execution Order
2023-01    41500
2023-02    33050
2023-03    29300
2023-04    42900
2023-05    34600
2023-06    34600
2023-07    52700
2023-08    46850
2023-09    35500
2023-10    33550
2023-11    55100
2023-12    35150
2024-01    58450
2024-02    51450
2024-03    33350
2024-04    30350
2024-05    39550
2024-06    35750
2024-07    49800
2024-08    36500
2024-09    32150
2024-10    34600
2024-11    17650
2024-12    32400
Freq: M, Name: Revenue, dtype: int64
```

8. Revenue Trend Analysis

A revenue trend chart was created to visualize **sales performance over time**. This allows us to observe:

- Growth patterns
- Seasonal fluctuations
- Potential business cycles

```
Montly_revenue.plot(
    kind="line",
    figsize=(10,5),
    title="Montly Revenue"
)

plt.xlabel("Month")
plt.ylabel("Revenue")
```



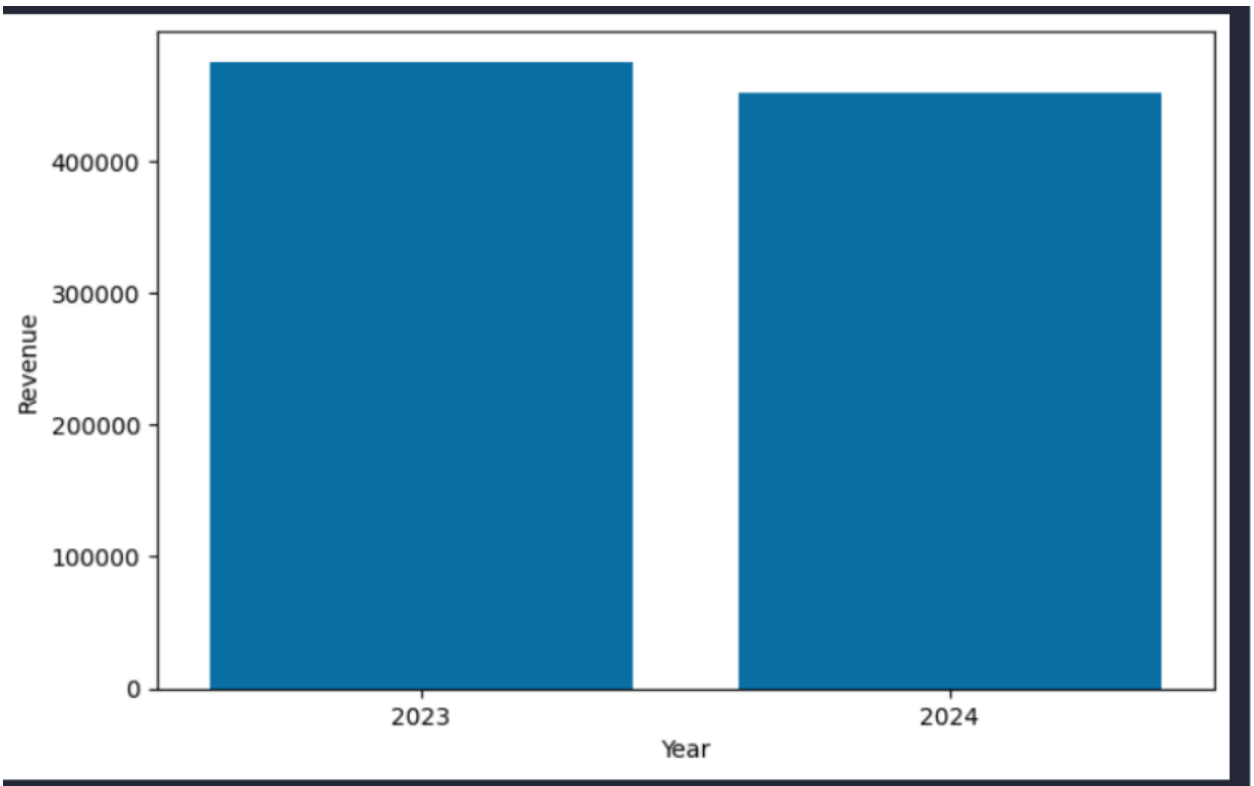
9. Revenue by Year

Insight

Total revenue remained relatively stable between **2023 and 2024**, with both years generating approximately **$2.3 million in sales.**

This suggests:

- Consistent customer demand
- A stable market position
- Limited year-over-year growth




```
Year  Revenue
2023   474800
2024   452000
```

Total revenue remained relatively stable between 2023 and 2024, with both years generating approximately $2.3M in sales. This suggests consistent demand but limited year-over-year growth.

10. Pareto Analysis

A Pareto analysis was conducted to determine whether **a small number of products generate the majority of revenue**.

This analysis follows the **80/20 principle**, which suggests that:

- Roughly 80% of results come from 20% of the causes.

```
pareto["Cumulative_revenue"] = pareto["Revenue"].cumsum()
pareto["Cumulative_percentage"] =pareto["Cumulative_revenue"]/ pareto["Revenue"].sum() *100

print(pareto)

✓ 0.1s

      Product  Revenue  Cumulative_revenue  Cumulative_percentage
0      Laptop   153350              153350              16.546180
1     Monitor   150550              303900              32.790246
2       Mouse   133900              437800              47.237808
3      Tablet   132700              570500              61.555891
4  Headphones   127350              697850              75.296720
5    Keyboard   124100              821950              88.686880
6  Smartphone   104850              926800             100.000000
```

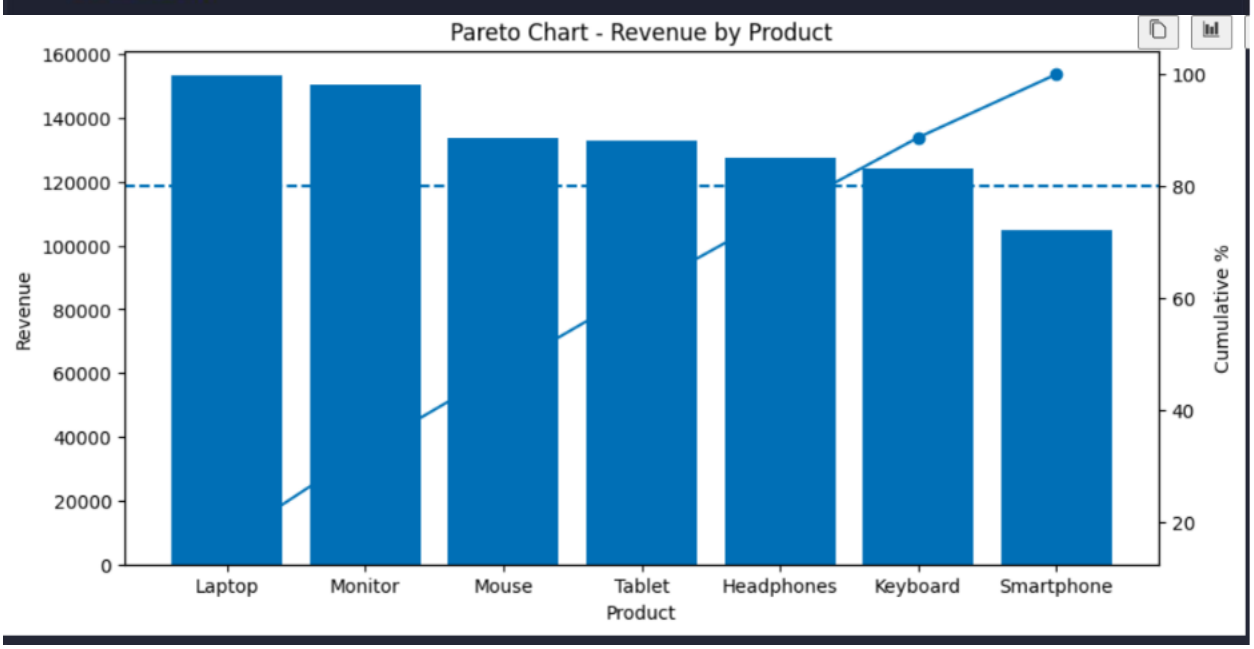
10. Pareto Chart

The Pareto chart visually shows:

- Individual product contributions
- Cumulative revenue percentage

This helps identify **which products have the greatest impact on revenue**.

```python
pareto = pareto.sort_values(by="Revenue", ascending=False)

fig, ax1 = plt.subplots(figsize=(10,5))


ax1.bar(pareto["Product"], pareto["Revenue"])
ax1.set_ylabel("Revenue")
ax1.set_xlabel("Product")


ax2 = ax1.twinx()
ax2.plot(pareto["Product"], pareto["Cumulative_percentage"], marker="o")
ax2.set_ylabel("Cumulative %")


ax2.axhline(80, linestyle="--")

plt.title("Pareto Chart - Revenue by Product")

plt.show()
```



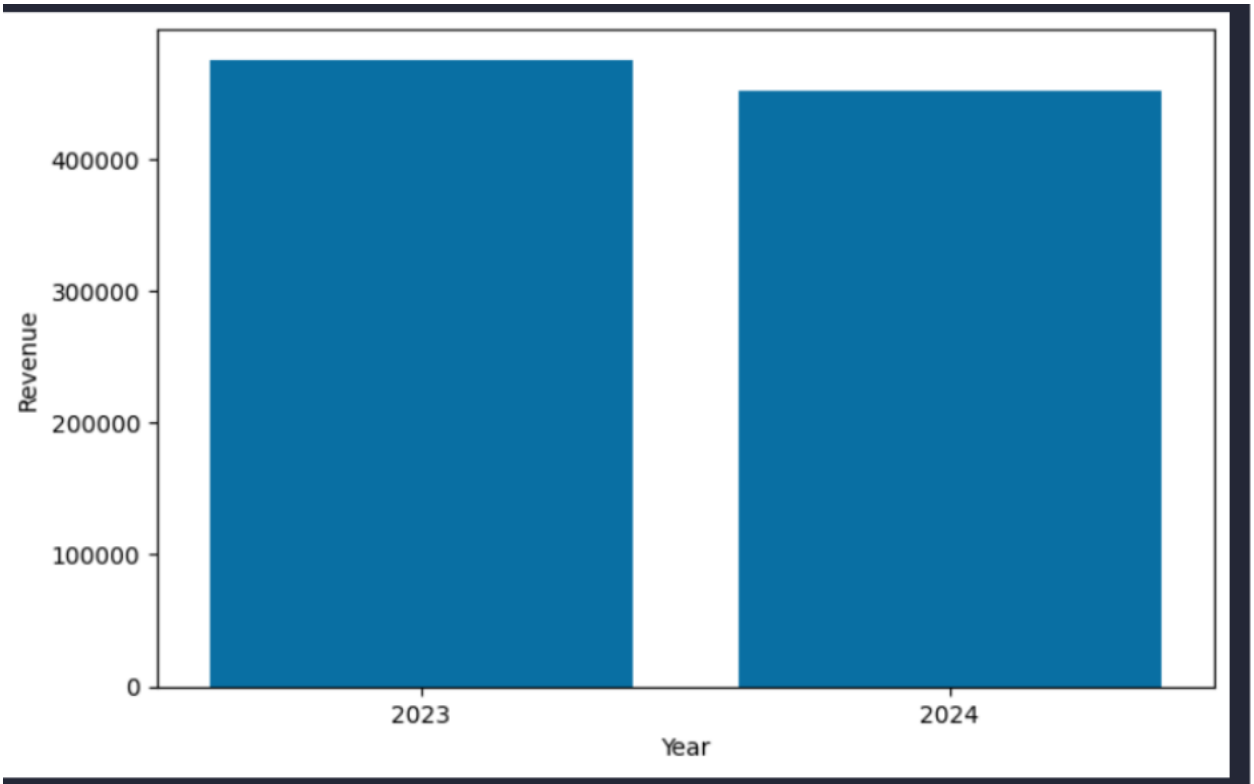
Business Insights and Recommendations

Overall Revenue Performance

The company generated strong revenue across the analyzed period. Total revenue remained **stable between 2023 and 2024**, with both years producing similar results.

While growth was limited, the stability suggests:

- A **consistent customer base**
- eliable product demand




```
Year  Revenue
2023   474800
2024   452000
```

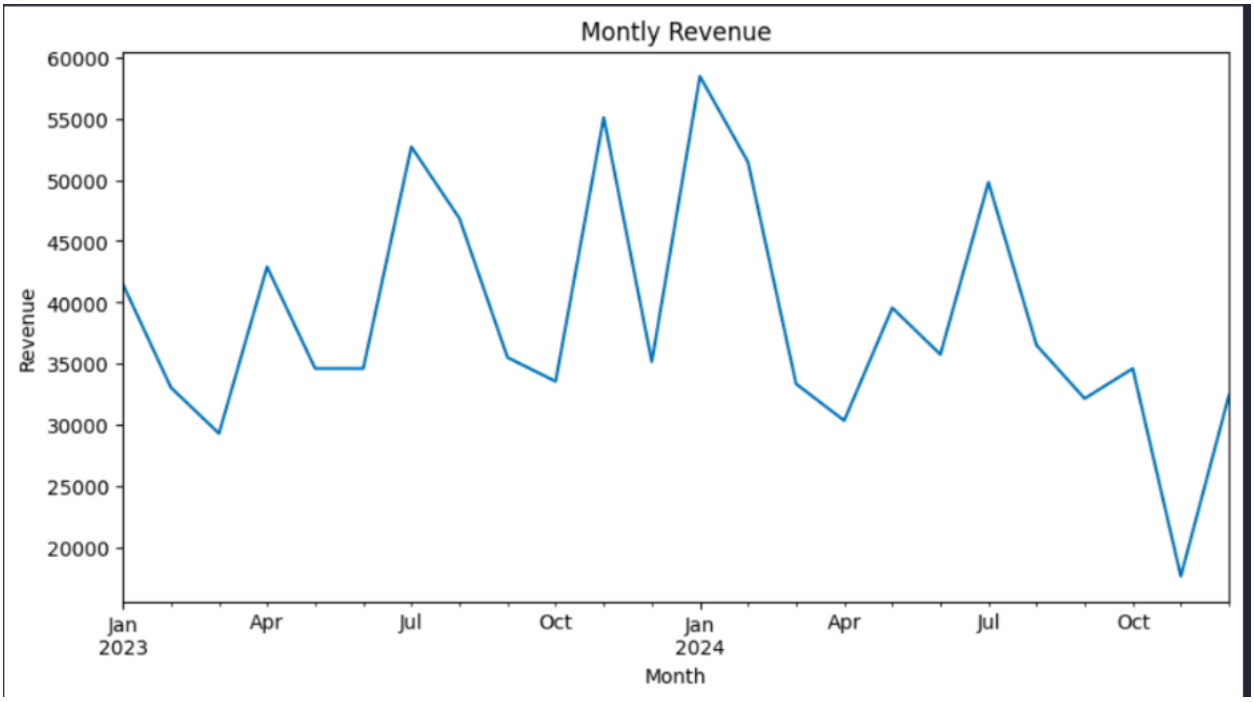
Seasonal and Monthly Sales Trends

The monthly trend analysis revealed **noticeable fluctuations throughout the year**, indicating seasonal purchasing behavior.

Certain months consistently generate higher revenue, which may correspond to:

- Promotional periods
- Holiday seasons
- Peak demand cycles

Understanding these patterns can help the company **better plan marketing campaigns and inventory levels**.



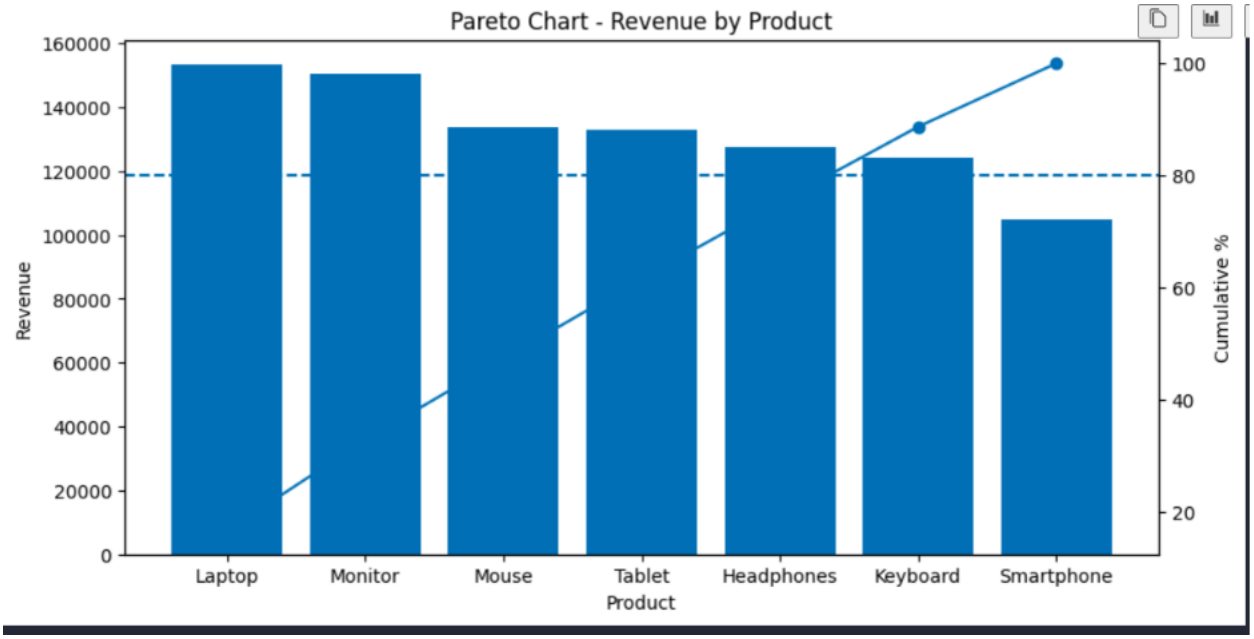
Product Contribution Analysis

The Pareto analysis shows that **a small number of products generate a large percentage of total revenue.**

This indicates that company revenue is **highly concentrated among top-performing products.**

Focusing on these key products can help increase profitability through:

- Targeted marketing
- Strategic inventory management
- Improved product availability



Regional Sales Performance

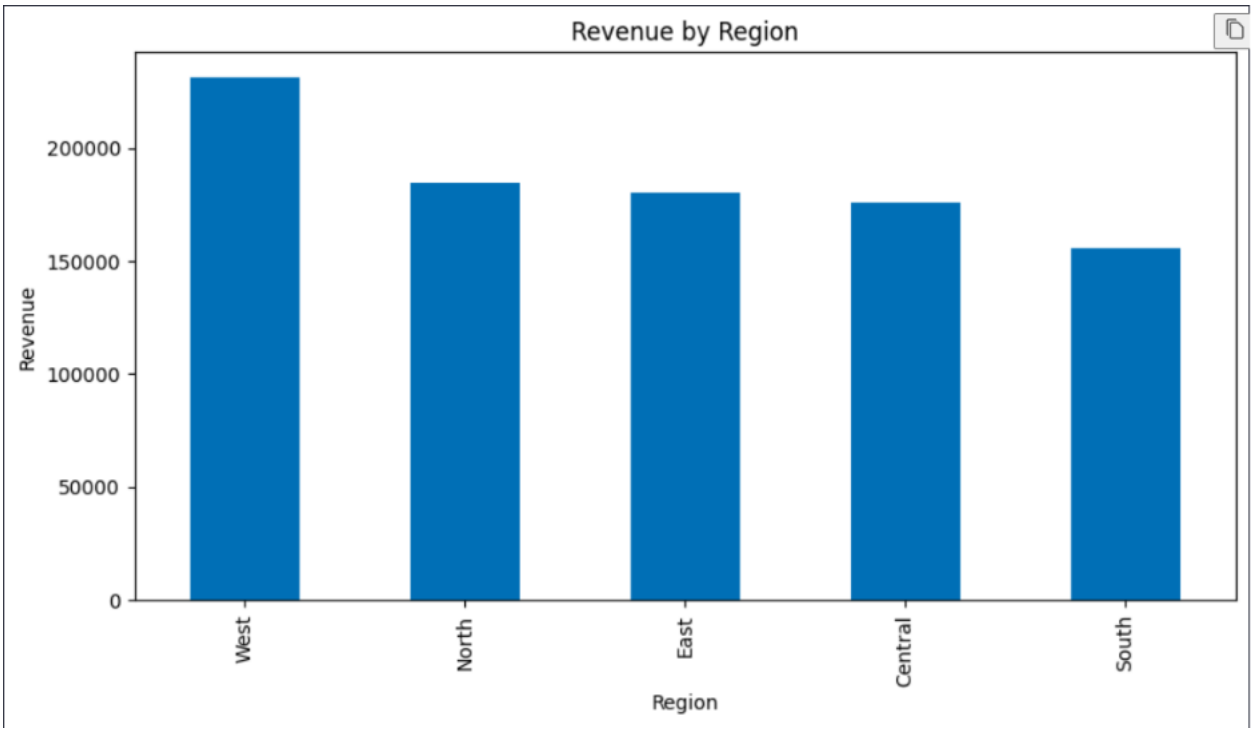
Regional analysis reveals that **certain regions contribute significantly more revenue than others.**

Major metropolitan areas generate the highest sales, suggesting:

- Stronger market penetration
- Higher customer demand

Meanwhile, regions with lower sales may represent **untapped growth opportunities**.

Expanding marketing efforts or improving distribution strategies in these areas could increase overall revenue.



Strategic Recommendations

Based on the analysis, several opportunities for business improvement were identified:

1. **Prioritize high-performing products** that generate the majority of revenue.
2. **Align marketing campaigns with seasonal demand peaks** to maximize sales during high-revenue months.
3. **Expand marketing and sales strategies in underperforming regions** to capture new market opportunities.
4. **Monitor year-over-year trends** to identify growth opportunities and potential revenue risks.